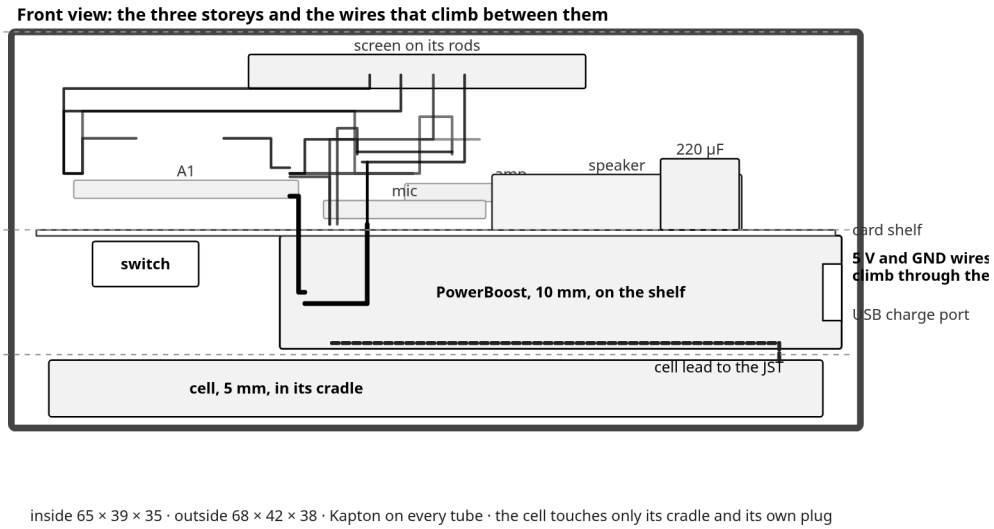
POCKET AI ASSISTANT · FINAL BUILD · EVERYTHING IN ONE FRAME

The whole build in one frame

Harness, PowerBoost, cell and switch, all inside one 68 × 42 × 38 frame, drawn storey by storey from the same model. Every rod of the harness is here unchanged (its own page has the per-rod drawings and cut lengths); this page adds the five connections that make it a battery device and shows where each thing physically sits.

The frame and its three storeys

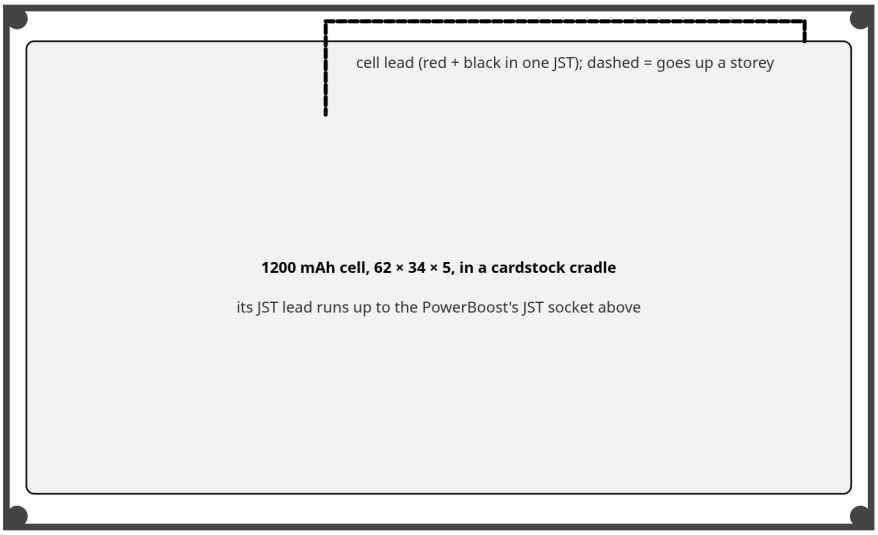


inside 65 × 39 × 35 · outside 68 × 42 × 38 · Kapton on every tube · the cell touches only its cradle and its own plug

From the bottom up: the cell in a cardstock cradle (6 mm), a card shelf carrying the PowerBoost flat with its USB port at the right opening (11 mm), and the harness standing on that shelf (16 mm), the screen floating on its rods. Two wires climb through holes in the shelf to A1's 5V and GND stubs. Total 33 of the 35 mm inside.

Storey 1: the cell

Storey 1, the bottom: the cell only

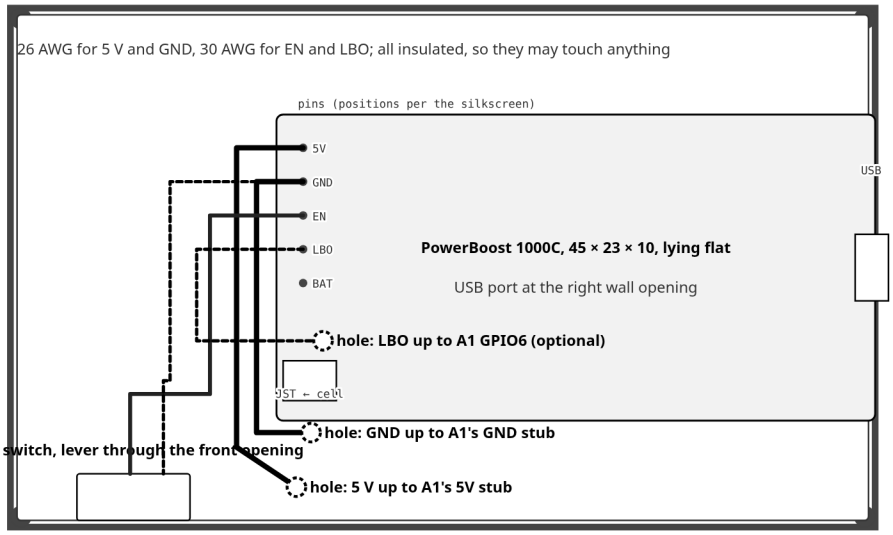


front of the device at the bottom · inside 65 × 39 · 10 px = 1 mm

Nothing touches the cell but its cradle. 62 × 34 mm of cardstock box, 1 mm of margin, the pouch flat, its own JST lead led up to the PowerBoost. No solder, no brass, no iron near it; it goes in last and comes out first.

Storey 2: the shelf, the PowerBoost, the switch

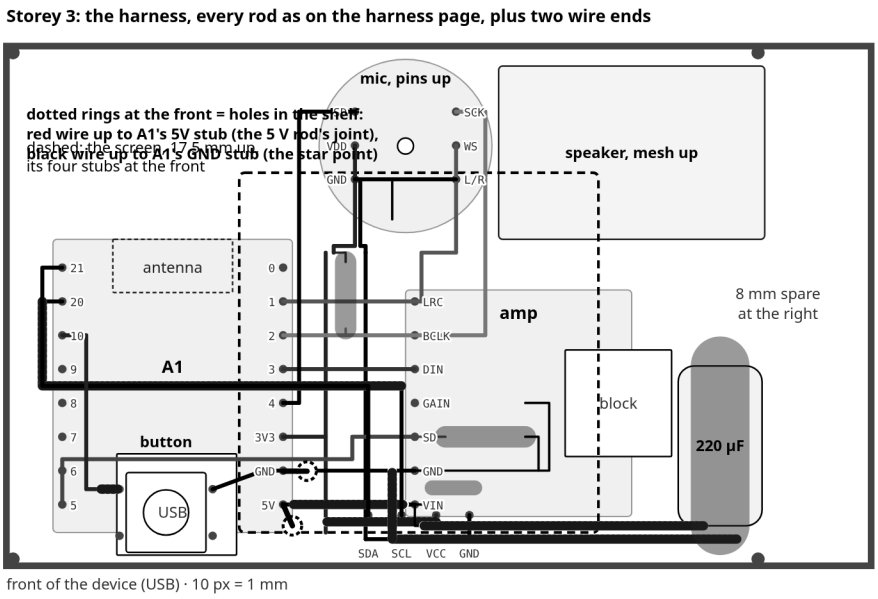
Storey 2: the card shelf, the PowerBoost, the switch, and the wires that go up



front of the device at the bottom · the harness stands on this shelf

Five wires leave the PowerBoost. 5 V (red) and GND (black) climb through two holes in the shelf to the harness; EN (purple) goes to the slide switch, whose other side returns to PowerBoost GND; LBO (teal, optional) climbs to A1's GPIO6. The cell's JST plugs into the socket at the board's left end. Being insulated wire, these may cross and touch anything.

Storey 3: the harness, with its two new joints



Every rod exactly as on the harness page, left-aligned in the wider frame with 8 mm spare on the right. The red wire from below ends on A1's 5V stub (the same stub the 5 V rod starts from); the black wire ends on the GND rail's front end. A1's USB socket gets a square of Kapton: it is no longer a power input.

Every connection in the device

Power and battery (new)

#	FROM	TO	WITH	NOTE
P1	PowerBoost `5V`	A1 5V stub (through the shelf hole at x = 21.5, y = 3)	26 AWG red, ≈ 60 mm, TUOFENG	joins the 5 V rod's joint; the 220 µF stays where it is
P2	PowerBoost `GND`	A1's GND stub (hole at x = 22.6, y = 7.1)	26 AWG black, ≈ 55 mm	the star point; the rail stays a dead end
P3	PowerBoost `EN`	slide switch, one side	30 AWG, ≈ 40 mm	the other side of the switch to PowerBoost `GND` (P4). Closed = EN grounded = off
P4	switch, other side	PowerBoost `GND`	30 AWG, ≈ 40 mm	
P5	1200 mAh cell JST	PowerBoost JST socket	the cell's own lead	check + and − against the board's silkscreen before the first plug-in
P6	PowerBoost `LBO`	A1 GPIO6	30 AWG, optional	low-battery flag for a later firmware face
–	A1 USB-C	nothing	Kapton over it	flashing only, switch off first

The harness (unchanged)

A1 PIN	GOES TO	BY	LENGTH
5V	amp VIN	5 V rod	10 mm
GND	amp GND, and the GND rail	GND rod + GND drop pin	14 mm
3V3	3V3 rail	3V3 tap	7 mm
GPIO4	mic SD	mic SD rod	32 mm
GPIO3	amp DIN	DIN rod	10 mm
GPIO2	amp BCLK, then mic SCK	BCLK rod	38 mm
GPIO1	amp LRC, then mic WS	WS rod	33 mm
GPIO5	amp SD	amp SD rod	43 mm
GPIO10	button	button rod	19 mm
GPIO20	screen SCL	SCL rod	53 mm
GPIO21	screen SDA	SDA rod	53 mm

TAP OR PART	BETWEEN
screen GND tap · screen VCC tap	screen GND → GND rail front · screen VCC → 3V3 rail front
mic GND tap · mic VDD tap · L/R link	mic GND → GND rail back · mic VDD → 3V3 rail back · mic L/R → mic GND
button GND rod	button right legs → A1 GND stub
R1 4.7 kΩ	amp SD → amp GND (GAIN left open, 9 dB)
R3 10 kΩ	BCLK rod → 3V3 rail, until A1's own GPIO2 pull-up is confirmed (mic SD pull-down is in firmware)
C1 220 µF · C2 100 nF	5 V net → GND rod · amp VIN → amp GND
speaker	OUT+ / OUT− only, never GND or the frame

Build order for the one-frame version

1. The harness, moves 1–11 of the harness page, on USB, on its jig card. Make it work first.
2. The frame to the new size: two loops at 65 × 39 inside, four posts of 35. Kapton inside.
3. The shelf: a 64 × 38 card with the two wire holes and the LBO hole punched where the storey-2 drawing shows; the switch glued at its front edge.
4. PowerBoost on the shelf, USB port at the right wall; wires P1–P4 (and P6) soldered at the PowerBoost end first, led up through the holes.
5. Harness off its jig and onto the shelf; P1 to A1's 5V stub, P2 to A1's GND stub. Kapton over A1's USB.
6. Cradle and cell at the bottom, JST into the PowerBoost. Meter frame to everything: silent.
7. Switch on: the same tests as move 11. Then charge it once and watch the LEDs.

Sources. The harness from `edu/pages/gen/harness_model.py`; PowerBoost 1000C dimensions, pins and functions from Adafruit's product page; cell size from `docs/INVENTORY.md`. The lower storeys are a layout proposal measured against those sizes, not a built thing; pin positions on the PowerBoost are drawn as a group and must be read off its silkscreen.